

Implementation Plan

AuraHR

Intelligent Workplace OS

A comprehensive solution architecture addressing HR management inefficiencies, onboarding, attendance, and performance tracking.

1. Problem Statement & Context

The modern HR ecosystem is highly fragmented. Employees and HR teams struggle with siloed platforms for **leave management, onboarding, attendance tracking, and performance workflows**. This fragmentation leads to inefficient onboarding experiences, communication gaps, slow HR helpdesk response times, and manual tracking errors.

The AuraHR Solution

AuraHR consolidates these disparate workflows into a unified, AI-driven platform. It ensures a frictionless employee experience from day one while significantly reducing the administrative burden on HR teams.

2. Solution Architecture & Workflow

Technical Stack

Layer	Technology & Rationale
Backend API	FastAPI (Python): Selected for high-performance, asynchronous API routing, ensuring rapid response times during peak check-in hours.
Database & Auth	Supabase: Utilized for PostgreSQL data storage, robust authentication, and Row-Level Security (RLS) to protect employee records.
AI HR Helpdesk	Ollama + Llama 3: A locally orchestrated LLM ensures that employee queries are processed instantly and securely, strictly maintaining data privacy without relying on external APIs.
Hosting & DevOps	Render & GitHub Codespaces: Services deployed via Render for seamless scaling, with development standardized through Codespaces.

Process Architecture

- **AI Onboarding:** Upon user creation in Supabase, an automated workflow provisions accounts and triggers the LLM to generate a personalized onboarding checklist and welcome packet.
- **Attendance Tracking:** Geofenced or IP-based check-ins log directly to the database. The system automatically reconciles these logs to calculate working hours.
- **Automated Helpdesk:** Employee queries (e.g., "What is our remote work policy?") are embedded and queried against the company's internal documents. Llama 3 synthesizes accurate, conversational responses instantly.
- **Performance Workflow:** Continuous feedback is logged via FastAPI endpoints. During review cycles, the AI summarizes this continuous data to aid managers in evaluations.

3. UI/UX Design System

To ensure a high-fidelity, professional user experience, the system adheres to a modern SaaS design language optimized for readability and reduced eye strain.

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- **Layout:** Fixed sidebar navigation for module switching. Main content utilizes a widget-based card dashboard for quick data consumption.
- **Typography:** Clean sans-serif (Inter/Roboto) ensuring high legibility for data-dense performance and attendance tables.
- **Feedback:** Immediate visual feedback using the success/warning color palette for leave approvals and attendance check-ins.

4. Project Rationale & Impact

Why this problem matters

HR inefficiencies cost organizations thousands of hours annually. A confusing onboarding process increases 90-day turnover rates, while slow helpdesk responses frustrate employees and bottleneck productivity.

Expected Business Impact

- **70% reduction** in routine HR ticket volume via the AI Helpdesk.
- **Accelerated Onboarding:** Centralized resources reduce time-to-productivity.
- **Zero-Friction Tracking:** Automated, error-free attendance and leave accrual workflows.